

Targeted Reduction of Glutamate Excitotoxicity and Oxidative Stress in Amyotrophic Lateral Sclerosis (ALS)

University of Connecticut, Storrs

Kelvin James, Jon Balyeat, Zayna Abraham, Jada Fox, Julia Armington

Submitted: May 6, 2026

Group 7

Application

Amyotrophic lateral sclerosis (ALS) is a progressive neurodegenerative disease that attacks motor neurons in the brain and spinal cord. These motor neurons control voluntary muscle movement. As they deteriorate, patients gradually lose their ability to move, speak, swallow, and eventually breathe [1]. ALS exists in two main forms. Around 90 to 95 percent of cases are classified as sporadic ALS and occur randomly without any family history. The remaining 5 percent are familial ALS, meaning the disease runs in families due to inherited genetic mutations in genes like SOD1 and C9orf72 [2]. Scientists have identified genetic factors, although the exact cause of ALS remains unknown. Many researchers believe the disease results from a combination of genetic vulnerabilities, environmental factors, and cellular dysfunctions. The progression of ALS is relentless and typically follows a pattern where patients first notice weakness in their limbs or difficulty speaking. Over time, the weakness spreads throughout the body until patients become completely paralyzed, while their cognitive function often remains intact. This means patients are usually fully aware of their declining physical state, which adds to the psychological burden of the disease. [3], [4]

The National ALS Registry reported that there were approximately 30,000 confirmed cases of ALS in the United States in 2018 [5]. Although a rare disease, it is accompanied by a poor prognosis; therefore, individuals with the disease typically do not survive longer than 5 years past diagnosis. This fatal disease progresses rapidly, leading to a fast decline in bodily function for affected individuals. Due to the impact ALS has on the spinal cord, brain, and neurons in the body, the most common cause of death with ALS is respiratory failure. This devastating outlook and no known cure to the disease give reason to increase ALS research to expand knowledge on the condition and possible treatment options. In its aggressive nature, reported cases of ALS have shown an increase and are projected to increase in the following years. Statistical projections show that cases are expected to increase 10% from 2022 to the year 2030, reaching an estimated 37,000 cases per year in the United States [6]. This prediction urges greater funding and development regarding ALS therapeutics to address the growing nature of the condition.

A major challenge in ALS is that motor neuron degeneration is driven by multiple interacting toxic mechanisms rather than a single isolated defect. Excess extracellular glutamate can overstimulate motor neurons and contribute to excitotoxic injury. Oxidative stress damages proteins, lipids, and DNA. In addition, abnormal protein aggregation, including TDP-43 pathology, and chronic neuroinflammation further disrupt the central nervous system microenvironment. These mechanisms suggest that future therapeutic strategies should not only focus on symptom management, but also on reducing the toxic factors that accelerate motor neuron dysfunction. Therefore, a key area of investigation is how toxic molecules and damaging cellular stressors in the CNS environment can be removed, neutralized, or controlled to preserve remaining motor neuron function. The goal of this project is to preserve remaining motor neuron function by locally reducing glutamate-mediated excitotoxicity and oxidative stress in the spinal CNS microenvironment.

Background

The primary focus of this literature search is understanding how current and emerging treatments address the cellular mechanisms driving motor neuron degeneration in ALS, specifically excitotoxicity and oxidative stress, rather than symptom management alone. Amyotrophic lateral sclerosis (ALS) has few clinical therapeutic options, and neither disease reversal nor restoration of lost motor neuron function is currently available. The two FDA-approved disease-modifying treatments, riluzole and edaravone, provide only modest benefits. Riluzole reduces glutamate-related excitotoxicity and extends survival by about two to three months, while edaravone may slow functional decline in some patients [7], [17]. However, both treatments are limited in their effectiveness since they do not significantly stop disease progression or reverse motor neuron damage. Most other care is supportive and multidisciplinary, including nutritional support, speech therapy, respiratory treatment, non-invasive ventilation, and medications for cramps, spasticity, and excessive salivation.

Recent tissue engineering and biomaterials research aims to address ALS beyond systemic pharmacological treatment. In a 2025 study, Rodriguez-Romano et al. used the SOD1G93A mouse model to test injectable borax-loaded alginate hydrogels [9]. This approach aligns with the “dying-back” hypothesis, in which degeneration progresses from distal neuromuscular structures toward motor neurons. In mice, the treatment improved motor function, reduced inflammation, and increased metabolic activity [9]. This demonstrates that injectable biomaterials are being explored for ALS treatment, but this approach remains preclinical and does not directly address central nervous system toxicity, limiting its ability to fully prevent motor neuron degeneration. Other nanostructured systems, lipid-based nanocarriers, and hydrogel-based localized delivery strategies have also been explored for neural applications because they offer enhanced control over drug delivery, molecular transport, and interactions with the cellular microenvironment [10]–[12], [25.5]. However, these systems still face challenges in reliable targeting, off-target effects, therapeutic stability, and controlled degradation.

Despite these advances, major translational and engineering limitations remain. Many experimental systems still struggle to maintain stable localization within the spinal environment, achieve sustained therapeutic activity, minimize off-target effects, and balance injectability with long-term biocompatibility and material retention in soft neural tissue [13]. These limitations suggest that an effective ALS biomaterial platform should provide localized, sustained delivery within the spinal microenvironment while reducing both glutamate-mediated excitotoxicity and oxidative stress to preserve surviving motor neurons. This project therefore focuses on developing an injectable hydrogel capsule system capable of locally delivering glutamate-removal and ROS-reduction components to reduce toxic stressors in the ALS spinal microenvironment.

Design

Although current pharmacological treatments, such as riluzole and edaravone, can modestly slow ALS progression, they do not directly address the elevated reactive oxygen species (ROS) and excess extracellular glutamate that contribute to neuron degradation. To address this limitation, we propose a novel biomaterial consisting of small dual-chamber hydrogel beads, which mimic the localized depots formed following multiple intraspinal microinjections. The chamber walls would primarily be composed of a low-swelling guest-host hyaluronic acid (HA) hydrogel made from adamantane-modified HA and β -cyclodextrin-modified HA, allowing reversible molecular interactions to provide shear-thinning and self-healing behavior during injection [21], [26]. One chamber will contain glutamate oxaloacetate transaminase (GOT), while the other will contain sulforaphane-loaded lipid nanoparticles. The GOT chamber would also include a mild Ca^{2+} -crosslinked alginate microgel interior to better protect the enzyme payload. The beads would be fabricated using extrusion-based bioprinting or microfabrication of a modified HA hydrogel ink, since HA-based bioinks can be engineered for shear-thinning, self-healing, and secondary crosslinking to maintain printed structure after fabrication [43], [44]. This method of bioprinting also allows for the beads to be produced on the micrometer scale. This allows for injection into the ventral horn, which has a diameter of up to 50 μm . Additionally, the small size of the beads maintains the ability to inject up to 5-12mL of the biomaterial so as not to increase intracranial pressure.

The proposed hydrogel's shear-thinning capabilities allow it to rapidly form a semi-stable hydrogel after placement. This localized retention is intended to help minimize rapid diffusion through cerebrospinal fluid and maintain therapeutic concentrations near damaged spinal tissue. In the glutamate-removal chamber, GOT catalyzes the conversion of excess extracellular glutamate and oxaloacetate into aspartate and α -ketoglutarate, reducing excitotoxic stress within the neural microenvironment [36], [37]. This chamber would contain GOT, pyridoxal phosphate (PLP), oxaloacetate, sucrose or trehalose, and a near-physiological HEPES buffer. PLP is required for GOT activity; oxaloacetate serves as the amino-group acceptor, and sucrose or trehalose helps stabilize the enzyme against denaturation during fabrication, storage, injection, and release. In the oxidative-stress chamber, sulforaphane-loaded lipid nanoparticles enable sustained antioxidant release while protecting the payload from premature degradation. Sulforaphane activates the Nrf2 antioxidant pathway, increasing downstream antioxidant-response genes such as HO-1 and NQO1, while lipid nanoparticles improve sulforaphane stability and release consistency [25.5], [38].

This design offers several advantages over existing ALS treatments and current experimental biomaterial systems. Unlike riluzole and edaravone, which act systemically and provide limited disease modification, the proposed hydrogel directly targets the damaged neural microenvironment. The combination of GOT-mediated glutamate reduction with sustained sulforaphane antioxidant delivery enables simultaneous targeting of two major pathological mechanisms using a single injectable platform. Also, the HA-based hydrogel has soft mechanical properties more like native neural tissue and supports localized retention within soft neural tissue [21], [29].

Characterization

There are several parameters that need to be tested for the injectable hydrogel capsule system to function properly. First, the expected lifetime of the device must be tested for the outer hyaluronic acid (HA)-based hydrogel carrier, the GOT-containing glutamate-removal chamber, and the sulforaphane-loaded lipid nanoparticle chamber. The hydrogel beads will be incubated at 37 °C in synthetic cerebrospinal fluid to simulate the spinal environment. Swelling ratio, percent mass remaining, and degradation behavior will be measured to ensure that the hydrogel remains localized and does not expand enough to compress spinal tissue [21], [29]. GOT activity will be tested by collecting surrounding solution over time and measuring glutamate concentration, since GOT converts glutamate and oxaloacetate into aspartate and α -ketoglutarate [36], [37]. The expectation is that extracellular glutamate will decrease over time if the enzyme chamber remains active. The sulforaphane-loaded lipid nanoparticles will be tested for particle size, polydispersity index, and release rate to confirm sustained antioxidant delivery and NRF2 pathway activation [25.5], [38]. The expectation is that sulforaphane will release gradually rather than all at once, allowing prolonged antioxidant pathway activation.

The next necessary parameter is mechanical and injection performance. Rheological testing will measure storage modulus (G'), loss modulus (G''), viscosity as a function of shear rate, and percent recovery after high-strain cycles [26], [27]. The hydrogel should behave as a soft solid at rest, with G' greater than G'' , but shear-thin during injection and rapidly recover after shear is removed. Injection testing will be performed by passing the hydrogel capsules through a fine cannula into a synthetic spinal tissue phantom. This will determine whether the beads maintain their structure, whether the chambers remain separated, and whether reflux or dispersion occurs. These tests are important because the material must survive injection while forming a localized depot near the spinal target region. The expected outcome is that the capsules can be injected with minimal rupture, minimal reflux, and limited spreading away from the intended depot region.

Chemical and structural characterization will verify that the hydrogel capsule has the intended composition and architecture. FTIR will confirm chemical features of the HA-based hydrogel, while Cryo-SEM will examine pore structure, chamber boundaries, and internal morphology without collapsing the hydrated network [28]. Output measures will include mean pore diameter, pore interconnectivity, and visible separation between the GOT and sulforaphane chambers. These structural features are important because the capsule must allow small molecules like glutamate and oxaloacetate to diffuse while retaining the larger enzyme and nanoparticle payloads. The system will be considered successful if it demonstrates shear-thinning, rapid self-healing, low swelling, gradual degradation, preserved GOT activity, sustained sulforaphane release, and stable dual-chamber structure after fabrication and injection [21], [25.5], [26]–[29], [36]–[38].

Biological Characterization

To evaluate safety and therapeutic potential, both **in vitro** and **in vivo** biological characterization will assess whether the injectable hydrogel system reduces glutamate-mediated excitotoxicity, oxidative stress, and neuronal damage in ALS [33]. For in vitro testing, the complete hydrogel capsule format will be used rather than a flat sheet or isolated nanoparticle. Small hydrogel beads/capsules will be placed in direct contact with NSC-34 motor neuron-like cells, which are commonly used for ALS screening, with human iPSC-derived motor neurons used for validation if available [34], [35]. Cells will be exposed to elevated glutamate and oxidative stress conditions selected through dose-response testing. Experimental groups will include healthy untreated cells, disease/toxic controls, blank hydrogel capsules, GOT-only capsules, sulforaphane-LNP-only capsules, and full dual-chamber capsules. Short-term toxicity and ROS assays will be performed over 24–72 hours, while sustained neuroprotective effects and release behavior will be evaluated over 7 days.

The main in vitro outcomes will determine whether each capsule component functions as intended. Extracellular glutamate will be measured using a glutamate assay or HPLC to confirm GOT activity, since GOT converts glutamate and oxaloacetate into aspartate and α -ketoglutarate [36], [37]. Intracellular ROS will be measured using DCFH-DA fluorometry, while Live/Dead staining and MTS assays will assess neuron viability and metabolic activity. To confirm sulforaphane-mediated Nrf2 activation, NRF2 nuclear translocation will be measured by western blot or immunostaining, and RT-qPCR will quantify antioxidant gene expression, including HO-1 and NQO1 [38].

For in vivo testing, SOD1 transgenic ALS mice will be used to assess safety and therapeutic efficacy [33], [39], [40]. The hydrogel will be delivered through stereotactic intraspinal microinjection into the lumbar ventral horn using multiple low-volume injections to form localized depots and reduce pressure-related damage. Functional outcomes will include rotarod testing, CatWalk XT gait analysis, grip strength, body weight tracking, and electromyography [39], [40]. Biocompatibility will be evaluated through histology of the injection site, with GFAP and Iba1 immunostaining used to quantify astrocyte and microglial activation, and ELISA or RT-qPCR used to measure inflammatory cytokines such as TNF- α and IL-1 β [41], [42]. Controls will include sham surgery without hydrogel, untreated SOD1 ALS mice, riluzole-treated positive controls, and healthy wild-type mice. Spinal cord tissue will also be analyzed for ventral horn motor neuron survival, oxidative stress markers, NRF2 activation, inflammation, and local glutamate levels to confirm whether functional changes match the intended biological mechanism [36], [38], [39].

Conclusion

ALS remains a devastating neurodegenerative disease with limited therapeutic options and no current cure capable of reversing motor neuron loss. Existing treatments such as riluzole and edaravone provide only modest benefits and fail to sufficiently address the toxic spinal microenvironment that contributes to the disease progression. Recent advances in biomaterials, nanostructured systems, and localized delivery platforms demonstrate growing interest in therapies that directly target excitotoxicity and oxidative stress within neural tissue. The proposed dual-chamber hydrogel system builds upon these advances by combining localized glutamate removal and antioxidant pathway activation within a single injectable platform. Through material characterization and biological testing, the project aims to evaluate whether localized reduction of glutamate-mediated excitotoxicity and oxidative stress can help preserve motor neuron function and improve therapeutic targeting in ALS.

References

- [1] "Amyotrophic lateral sclerosis (ALS) - Symptoms and causes," Mayo Clinic. Accessed: Feb. 05, 2026. [Online]. Available: <https://www.mayoclinic.org/diseases-conditions/amyotrophic-lateral-sclerosis/symptoms-causes/syc-20354022>
- [2] "Amyotrophic Lateral Sclerosis (ALS) Treatment | UConn Health." Accessed: Feb. 05, 2026. [Online]. Available: <https://www.uconnhealth.org/neurology/services-specialties/amyotrophic-lateral-sclerosis>
- [3] "Amyotrophic Lateral Sclerosis (ALS) | National Institute of Neurological Disorders and Stroke." Accessed: Feb. 05, 2026. [Online]. Available: <https://www.ninds.nih.gov/health-information/disorders/amyotrophic-lateral-sclerosis-als>
- [4] "Amyotrophic lateral sclerosis: MedlinePlus Genetics." Accessed: Feb. 05, 2026. [Online]. Available: <https://medlineplus.gov/genetics/condition/amyotrophic-lateral-sclerosis/>
- [5] P. Mehta et al., "Prevalence of amyotrophic lateral sclerosis in the United States, 2018," *Amyotroph. Lateral Scler. Front. Degener.*, vol. 24, no. 7–8, pp. 702–708, Oct. 2023, doi: 10.1080/21678421.2023.2245858.
- [6] P. Mehta et al., "Amyotrophic lateral sclerosis estimated prevalence cases from 2022 to 2030, data from the national ALS Registry," *Amyotroph. Lateral Scler. Front. Degener.*, vol. 26, no. 3–4, pp. 290–295, Apr. 2025, doi: 10.1080/21678421.2024.2447919.
- [7] M. C. Kiernan et al., "Amyotrophic lateral sclerosis," *The Lancet*, vol. 377, no. 9769, pp. 942–955, Mar. 2011, doi: 10.1016/S0140-6736(10)61156-7.
- [8] "Amyotrophic Lateral Sclerosis (ALS) | National Institute of Neurological Disorders and Stroke." Accessed: Feb. 05, 2026. [Online]. Available: <https://www.ninds.nih.gov/health-information/disorders/amyotrophic-lateral-sclerosis-als>

- [9] A. Rodriguez-Romano, J. Gonzalez-Valdivieso, L. Moreno-Martinez, J. F. Vázquez Costa, R. Osta, and P. Rico, "Injectable borax-loaded alginate hydrogels reduce muscle atrophy, modulate inflammation, and promote neuroprotection in the SOD1G93A mouse model of ALS through mechanisms involving IGF-Akt-mTOR signaling," *Int. J. Biol. Macromol.*, vol. 319, no. Pt 4, p. 145645, Aug. 2025, doi: [10.1016/j.ijbiomac.2025.145645](https://doi.org/10.1016/j.ijbiomac.2025.145645).
- [10] Y. Han et al., "Towards 3D Bioprinted Spinal Cord Organoids," *Int. J. Mol. Sci.*, vol. 23, no. 10, p. 5788, May 2022, doi: [10.3390/ijms23105788](https://doi.org/10.3390/ijms23105788).
- [11] M. I. Prasanth, A. R. Mallya, W. C. Cho, and D. Mundekkad, "Nano-structured strategies in combatting neurodegeneration," *Front. Bioeng. Biotechnol.*, vol. 13, p. 1638668, Dec. 2025, doi: [10.3389/fbioe.2025.1638668](https://doi.org/10.3389/fbioe.2025.1638668).
- [12] A. Kwiatkowska, A. Grzeczkwicz, A. Lipko, B. Kazimierczak, and L. H. Granicka, "Emerging Approaches to Mitigate Neural Cell Degeneration with Nanoparticles-Enhanced Polyelectrolyte Systems," *Membranes*, vol. 15, no. 10, p. 313, Oct. 2025, doi: [10.3390/membranes15100313](https://doi.org/10.3390/membranes15100313).
- [13] L. Mazzini et al., "Results from Phase I Clinical Trial with Intraspinal Injection of Neural Stem Cells in Amyotrophic Lateral Sclerosis: A Long-Term Outcome," *Stem Cells Transl. Med.*, vol. 8, no. 9, pp. 887–897, Sep. 2019, doi: [10.1002/sctm.18-0154](https://doi.org/10.1002/sctm.18-0154).
- [14] M. E. Cudkowicz et al., "A randomized placebo-controlled phase 3 study of mesenchymal stem cells induced to secrete high levels of neurotrophic factors in amyotrophic lateral sclerosis," *Muscle Nerve*, vol. 65, no. 3, pp. 291–302, Mar. 2022, doi: [10.1002/mus.27472](https://doi.org/10.1002/mus.27472).
- [15] Wang, Z., Huang, J., & Yun, D. (2026). Current and emerging therapeutic strategies for amyotrophic lateral sclerosis: From pharmacological approaches to gene and stem cell therapies. *Frontiers in Neurology*, 17, Article 1729302. <https://doi.org/10.3389/fneur.2026.1729302>

- [16]. Gupta, D., Tator, C. H., & Shoichet, M. S. (2006). Fast-gelling injectable blend of hyaluronan and methylcellulose for intrathecal, localized delivery to the injured spinal cord. *Biomaterials*, 27(11), 2370–2379. <https://doi.org/10.1016/j.biomaterials.2005.11.015>
- [17]. Ursini, O., Grieco, M., Sappino, C., Capodilupo, A. L., Giannitelli, S. M., Mauri, E., Bucciarelli, A., Coricciati, C., de Turris, V., Gigli, G., Moroni, L., & Cortese, B. (2023). Modulation of methacrylated hyaluronic acid hydrogels enables their use as 3D cultured model. *Gels*, 9(10), Article 801. <https://doi.org/10.3390/gels9100801>
- [18]. Wanasingha, N., Dorishetty, P., Dutta, N. K., & Roy Choudhury, N. (2021). Polyelectrolyte gels: Fundamentals, fabrication and applications. *Gels*, 7(3), Article 148. <https://doi.org/10.3390/gels7030148>
- [19]. Shultz, R. B., & Zhong, Y. (2021). Hydrogel-based local drug delivery strategies for spinal cord repair. *Neural Regeneration Research*, 16(2), 247–253. <https://doi.org/10.4103/1673-5374.290882>
- [20]. Serra, M., Casas, A., Toubarro, D., Barros, A. N., & Teixeira, J. A. (2023). Microbial hyaluronic acid production: A review. *Molecules*, 28(5), 2084. <https://doi.org/10.3390/molecules28052084>
- [21]. Kang, D. Y., Cho, S., Lee, S. J., Kim, Y. C., & Kim, S. B. (2010). Extraction of hyaluronic acid (HA) from rooster comb and characterization using flow field-flow fractionation (FIFFF) coupled with multiangle light scattering (MALS). *Carbohydrate Polymers*, 81(4), 755–760.
- [22]. Shikina, E. V., Kovalevsky, R. A., Shirkovskaya, A. I., & Toukach, P. V. (2022). Prospective bacterial and fungal sources of hyaluronic acid: A review. *Computational and Structural Biotechnology Journal*, 20, 6214–6236.
- [23]. Nasatto, P. L., Pignon, F., Silveira, J. L. M., Duarte, M. E. R., Nosedá, M. D., & Rinaudo, M. (2015). Methylcellulose, a cellulose derivative with original physical properties and extended applications. *Polymers*, 7(5), 777–803. <https://doi.org/10.3390/polym7050777>
- [24] M. Danaei et al., “Impact of Particle Size and Polydispersity Index on the Clinical Applications of Lipidic Nanocarrier Systems,” *Pharmaceutics*, vol. 10, no. 2, p. 57, May 2018, Accessed: May 06, 2026. [Online]. Available: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC6027495/>
- [25]. M. H. Chen, L. L. Wang, J. J. Chung, Y.-H. Kim, P. Atluri, and J. A. Burdick, “Methods To Assess Shear-Thinning Hydrogels for Application As Injectable Biomaterials,” *ACS Biomater. Sci. Eng.*, vol. 3, no. 12, pp. 3146–3160, Dec. 2017, doi: 10.1021/acsbiomaterials.7b00734.

- [27]. A. Gefen, D. Weihs, A. Fremau, Y. V. D. Eynde, and E. Torfs, "Rheological Assessment for Determining Form Stability of Wound Dressings," *International Wound Journal*, vol. 22, no. 8, p. e70720, Aug. 2025, doi: 10.1111/iwj.70720.
- [28]. A. Velazco et al., "Reduction of SEM charging artefacts in native cryogenic biological samples," *Nat Commun*, vol. 16, no. 1, p. 5204, Jun. 2025, doi: 10.1038/s41467-025-60545-3.
- [29] M. Ciuro, M. Sangiorgio, G. Leanza, and R. Gulino, "A Meta-Analysis Study of SOD1-Mutant Mouse Models of ALS to Analyse the Determinants of Disease Onset and Progression," *Int. J. Mol. Sci.*, vol. 24, no. 1, p. 216, Dec. 2022, doi: 10.3390/ijms24010216.
- [30] L. Zhou, Q. Chen, J. Chen, and X. Li, "In Vitro Models of Amyotrophic Lateral Sclerosis," *International Journal of Molecular Sciences*, vol. 24, no. 20, p. 15362, 2023, doi: 10.3390/ijms242015362.
- [31] B. M. Hounoum et al., "NSC-34 Motor Neuron-Like Cells Are Unsuitable as Experimental Model for Glutamate-Mediated Excitotoxicity," *Frontiers in Cellular Neuroscience*, vol. 10, p. 118, 2016, doi: 10.3389/fncel.2016.00118
- [32] F. Campos et al., "Neuroprotection by glutamate oxaloacetate transaminase in ischemic stroke: An experimental study," *Journal of Cerebral Blood Flow & Metabolism*, vol. 31, no. 6, pp. 1378–1386, 2011, doi: 10.1038/jcbfm.2011.3..
- [33] Reactome, "oxaloacetate + glutamate $\rightleftharpoons$ aspartate + alpha-ketoglutarate [GOT1]," *Reactome Pathway Database*.
- [34] S. Sandouka and T. Shekh-Ahmad, "Induction of the Nrf2 Pathway by Sulforaphane Is Neuroprotective in a Rat Temporal Lobe Epilepsy Model," *Antioxidants*, vol. 10, no. 11, p. 1702, 2021, doi: 10.3390/antiox10111702.
Use this for sulforaphane activating the Nrf2 pathway, reducing oxidative stress, and supporting use of Nrf2-related markers such as HO-1/NQO1.
- [35] E. J. Bennett, R. J. Mead, M. Azzouz, P. J. Shaw, and A. J. Grierson, "Early Detection of Motor Dysfunction in the SOD1G93A Mouse Model of Amyotrophic Lateral Sclerosis (ALS) Using Home Cage Running Wheels," *PLOS ONE*, vol. 9, no. 9, p. e107918, 2014, doi: 10.1371/journal.pone.0107918.
- [36] S. Oliván et al., "Comparative study of behavioural tests in the SOD1G93A mouse model of amyotrophic lateral sclerosis," *Experimental Animals*, vol. 63, no. 2, pp. 147–153, 2014, doi: 10.1538/expanim.63.147.

[37] T. Philips and W. Robberecht, "Neuroinflammation in amyotrophic lateral sclerosis: role of glial activation in motor neuron disease," *The Lancet Neurology*, vol. 10, no. 3, pp. 253–263, Mar. 2011, doi: [https://doi.org/10.1016/s1474-4422\(11\)70015-1](https://doi.org/10.1016/s1474-4422(11)70015-1).

[38] W. Zhao, D. R. Beers, and S. H. Appel, "Immune-mediated Mechanisms in the Pathoprogession of Amyotrophic Lateral Sclerosis," *Journal of Neuroimmune Pharmacology*, vol. 8, no. 4, pp. 888–899, Jul. 2013, doi: <https://doi.org/10.1007/s11481-013-9489-x>.

[39] K. Nagaraja et al., "3D Bioprintable Self-Healing Hyaluronic Acid Hydrogel with Antimicrobial Activity," *Gels*, vol. 10, no. 12, p. 780, 2024, doi: 10.3390/gels10120780.

[40] L. Ouyang et al., "3D Printing of Shear-Thinning Hyaluronic Acid Hydrogels with Secondary Cross-Linking," *ACS Biomaterials Science & Engineering*, 2016, doi: 10.1021/acsbiomaterials.6b00158.